Basic information for winner announcement

Please provide your preferred information for use in announcing the winners of the competition.

- Name: **Dinh-Viet-Toan LE**
- Hometown: Lille, France
- A recent picture of yourself or digital avatar
- Social handle or URL (optional): https://dinhviettoanle.github.io/

- Name: **Samuel ANSEL**
- Hometown: Lille, France
- A recent picture of yourself or digital avatar
- Social handle or URL (optional): https://www.linkedin.com/in/samuel-ansel/

Model documentation and write-up

If you are a competition finalist, please answer the model documentation and write-up questions that were sent to you. Questions for your competition may differ from this set of example questions. Information included in this section may be shared publicly with challenge results. You can respond to these questions in an e-mail or as an attached file. Please number your responses.

1. Who are you (mini-bio) and what do you do professionally? If you are on a team, please complete this block for each member of the team.

- ***Samuel ANSEL***

I work at the Lille University Hospital (CHU de Lille, France) as a Research Engineer / Data Scientist, focusing primarily on textual healthcare data, where we leverage LLMs to improve patient care.

Since my engineering studies at ENSICAEN, where I followed a machine learning track, I spent 6 years working for a company in the field of Intelligent Document Processing before taking on my current role.

I have a strong passion for research, particularly in the healthcare field, where I believe my work can be genuinely beneficial to patients.

- ***Dinh-Viet-Toan LE***

I am a research engineer working in the health data warehouse of the Lille University Hospital (CHU de Lille, France), working on Natural Language Processing for healthcare applications.

I graduated from ISAE-SUPAERO (Toulouse, France) as an aeronautical engineer with a specialization in data science, followed by a PhD in Music Information Retrieval at University of Lille defended in November 2025.

2. What motivated you to compete in this challenge?

From a technical standpoint, while our daily research projects mainly focus on applying NLP techniques to analyze patients' textual documents, this data challenge gave us the opportunity to extend our skills beyond NLP, in particular in image processing. This also give us a way to explore new healthcare applications relying on artificial intelligence tools.

From a scientific and medical standpoint, although our core background is in data science rather than medicine, exploring data-driven approaches to early neurodegenerative disease detection is a rewarding application of our work. Our participation to this challenge may further open new research collaborations and exchanges between us, as data scientist, and medical specialists, either from imaging services from our hospital, or from external institutions such as the SFMN.

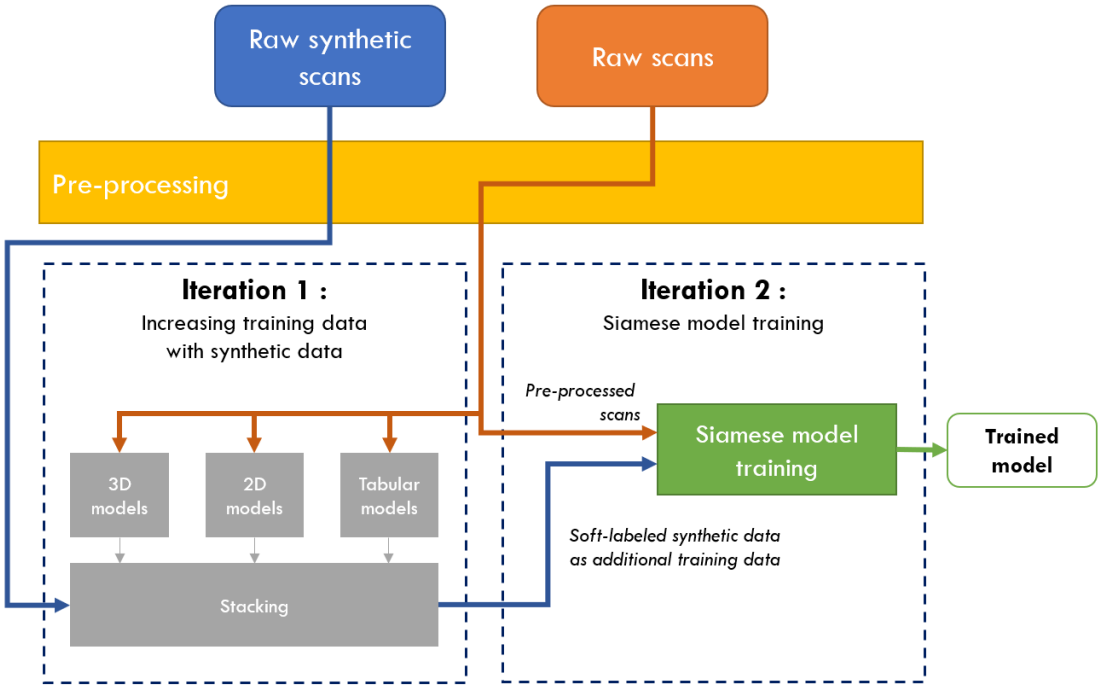
3. High level summary of your approach: what did you do and why?

We focus on the asymmetry between the left and right parts of the striatum, which represents the main geometric anomaly associated with Parkinsonian syndromes according to the provided problem description and a quick literature review.

To this end, the data processing pipeline is designed to isolate the striatum through a tight crop of the scans, followed by its separation into left and right components. This data is then passed through a Siamese neural network which further captures the symmetry or asymmetry of the striatum.

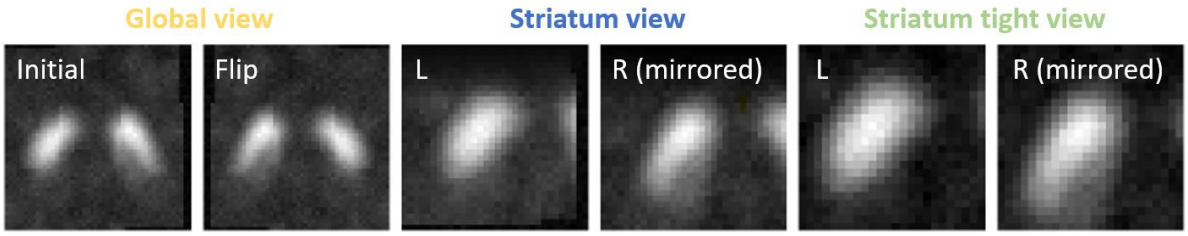
4. Do you have any useful charts, graphs, or visualizations from the process?

The full process relies on two blocks: (1) synthetic data soft-labeling which is then used as supplementary training data to (2) train a Siamese model to predict the actual scans.



5. Copy and paste the 3 most impactful parts of your code and explain what each does and how it helped your model.

- **Isolating the signal:** We found that most of the signal lies in the striatum, specifically in the axial plan, so we preprocessed the data to isolate each side of the striatum.
- **Siamese architecture:** We use a multi-view Siamese architecture that exploits striatum symmetry. We feed it with a global view of the striatum, one with each side independent and one close up of each side as represented below. Each view has a small dedicated CNN encoder.



- **Synthetic data:** The dataset is relatively small. In addition to cross validation techniques to take advantage of all the available data, we added soft labelled synthetic data [Won et al., 2025] that helped regularize the training process. While the real data (SPECT DaT scans) and the synthetic data (PET DaT scans) do not rely on the same technique (photon vs. positron emission), it appears that such augmentation helps the model training.

6. Please provide the machine specs and time you used to run your model.

- CPU (model): Intel® Xeon® 6517P @ 3.20GHz (16 cores)
- GPU (model or N/A): 2 x RTX PRO 6000 Blackwell Server Edition (VRAM: 2 x 96GB)
- Memory (GB): 512GB RAM
- OS: Ubuntu 26.04 LTS
- Python : 3.12.13
- Train duration:
    - Iteration 1 (synthetic data soft labeling): 3.5 days
        - Model trainings: estimated 3.5 days for all twelve models sequentially (about 7h per model)
        - Synthetic data soft-labeling (1500 samples): about 1h
    - Iteration 2 (Siamese model training): about 30 minutes
- Inference duration: 1h to 1h30 (for 1362 samples)

7. Anything we should watch out for or be aware of in using your model (e.g. code quirks, memory requirements, numerical stability issues, etc.)?

We found that predictions can be noisy on new data with models trained on such a small dataset. We tried to mitigate that with ensemble method like averaging prediction from 100 slightly different models. We also found that calibration with temperature scaling is needed to adjust model prediction on a new cohort. A few post-submission minor errors (such as data augmentation details) that we discover after the end of the competition and that we kept in the submitted solution have been documented in the README.

8. Did you use any tools for data preparation or exploratory data analysis that aren't listed in your code submission?

Python notebooks and usual visualization libraries we used for data preparation and preliminary exploration of the data. Neuroimaging viewers (e.g. VSCode's extension NeuroViewer) have also been used to visualize the raw scans.

9. How did you evaluate performance of the model other than the provided metric, if at all?

We used Log Loss and AUROC in a cross-validation setup. On our final iteration, each patient was evaluated with 20 models of a given architecture, where each of these 20 models have been but trained on different sets to reduce noise.

10. What are some other things you tried that didn't necessarily make it into the final workflow (quick overview)?

- Architectures and data:

    - Stacking as the final model: instead of using a unique Siamese model, stacking multiple types of models.
    - 2D-multiview: using sagittal, coronal and axial views as inputs, rather than only axial.
    - Discrete wavelet transform as an additional input view.
    - Re-labeling: modifying the labels of the samples from the training dataset which might be misannotated, according to the consensus between all our models.
    - Morphological features: we used computer vision algorithm to isolate the striatum, segment each side and head and tail. Then we compute features such as areas and intensity for each element (tail/head, L/R) and differences and ratios between each elements.
    - Striatum modelisation: we tried to model each side of the striatum with a 'spine' running from the head to the tail, sampling intensity along the spine. Then we trained a recurrent neural network which took the intensity serie of each side, encode them and process them like our CNN siamese network.
- Post-processing:
    - Projection: improving the log loss by clamping the prediction probability of a sample near to 0 or 1 according to its projection in a relevant space.

11. If you were to continue working on this problem for the next year, what methods or techniques might you try in order to build on your work so far? Are there other fields or features you felt would have been very helpful to have?

This question can be approached from two perspectives: improving our current models and exploring the new research directions that this challenge may open up.

On the one hand, as incorporating synthetic data have shown to improve our model's performances, we wonder whether the quality, diversity, and clinical relevance of the synthetic data (e.g. through conditioned generation of DaT scans) could be further improved to better capture the variability of the target population and potentially enhance model generalization. Additionally, incorporating further medical priors may help guide the model toward clinically relevant features. Investigating disease-related information beyond the striatum, whether across other anatomical regions or complementary imaging modalities, may provide complementary information and improve predictive performance.

On the other hand, this data challenge, seen as a research question, can open many future directions. First, the data provided in this competition only capture the patient's condition at a specific point in time. Since neurodegenerative diseases such as Parkinson's disease evolve over time, modeling the longitudinal evolution of scan characteristics, for instance through sequential models, could provide valuable insights for the clinical follow-up of patients. Moreover, real-world exams of Parkinson's disease are not based on imaging data alone. Brain scans are typically interpreted alongside clinical examination reports, biological measurements, and other patient-related information. Integrating imaging data within a multimodal framework that combines both textual and structured data could therefore provide a more comprehensive representation of the patient's condition and potentially improve the characterization and detection of Parkinsonian syndromes.

12. What simplifications could be made to run your solution faster without sacrificing significant accuracy?

As described in Q.7, we selected an arbitrary number of 20 runs trained on different shuffles of the 5-fold cross-validation splits to reduce prediction noise, resulting in an ensemble of 100 models for a single prediction. We hypothesize that a smaller number of models may be sufficient to achieve comparable performance while reducing computational cost.

Moreover, the first stage of our pipeline, which aims to generate soft labels for the synthetic data used to train the Siamese model, was initially developed as one of our first submissions and relied on a stack of several potentially large models. With the knowledge gained throughout the challenge, we believe that simpler and more computationally efficient models could be explored for this specific step.